

There are six problems in total. You must solve the first five, and problem 6 is optional.

Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language.

$$L = \{0^n 1^m : n = 3q + r_1, m = 3q + r_2, q \geq 0 \text{ and } 0 \leq r_1, r_2 \leq 2\}$$

Prove that L is not regular using the pumping lemma.

Problem 2 (CO3): Context-Free Grammars (15 points)

Consider the following languages.

$$\begin{aligned} L_1 &= \{w \in \{a, b\}^* : w \text{ is a palindrome}\} \\ L_2 &= \{w \in \{a, b\}^* : \text{every second letter in } w \text{ is } b\} \\ L_3 &= \{w \in \{a, b\}^* : w \text{ contains more } a \text{ than } b\} \\ L_4 &= L_1 \cap L_2 \\ L_5 &= L_2 \cap L_3 \end{aligned}$$

Now solve the following problems.

- (a) **Give** a context-free grammar that generates $\bar{L}_1$. (3 points) [Recall: $\bar{L}_1$ denotes the complement of the language L_1 i.e., $\bar{L}_1 = \Sigma^* - L_1$]
- (b) **Give** a context-free grammar that generates L_2 . (3 points)
- (c) **Give** a context-free grammar that generates L_3 . (3 points)
- (d) **Write** all five-letter strings in L_4 . (1 point)
- (e) **Give** a context-free grammar that generates L_4 . (2 points)
- (f) **Give** a context-free grammar that generates L_5 . (2 points)
- (g) **Find** the total number of strings that belong to $L_5 \setminus L_1$. (1 point) [Recall: $L_5 \setminus L_1$ contains all strings that are in L_5 but not in L_1 .]

Problem 3 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$\begin{aligned} S &\rightarrow P0Q \\ P &\rightarrow 0P0 \mid 0P1 \mid 1P0 \mid 1P1 \mid 0 \mid 1 \\ Q &\rightarrow 0Q1 \mid 1Q0 \mid \varepsilon \end{aligned}$$

- (a) **Give** a leftmost derivation for the string '1100101010'. (3 points)
- (b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)
- (d) **Write** two strings of length four that have exactly one parse tree in the given grammar. (1 point)

Problem 4 (CO3): Pushdown Automata (10 points)

Consider the following languages.

$$L_1 = \{w \in \{0,1\}^* : w \text{ starts with } 0 \text{ and ends with } 1\}$$

$$L_2 = \{w = 0^i 1^j 2^k : i, j, k \geq 0 \text{ and } k = 2i + j\}$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and the number of 0s in } w_1 \text{ is equal to the number of 1s in } w_2\}$$

Now solve the following problems.

- (a) **Give** the state diagram of a PDA that recognizes L_1 . (3 points)
- (b) **Give** the state diagram of a PDA that recognizes L_2 . (3 points)
- (c) **Give** the state diagram of a PDA that recognizes L_3 . (4 points)

Problem 5 (CO4): Turing Machines (10 points)

Consider the following languages.

$$L_1 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and } \text{count}(w_1, 0) = \text{count}(w_2, 0)\}$$

$$L_2 = \{w \# w' : w \in \{0,1\}^*\}$$

- (a) For the configuration $011q_4010$ of a Turing Machine, **what** is the current position of the tape head? (1 point)
- (b) **Give** the state diagram of a turing machine that decides L_1 . [Note: $\text{count}(w, a)$ denotes the number of occurrences of the symbol a in the string w .] (3 points)
- (c) **Give** the state diagram of a turing machine that decides L_2 . [Note: w' denotes the bitwise complement of w (obtained by flipping each bit: $0 \mapsto 1, 1 \mapsto 0$), for example, if $w = 01101$, then $w' = 10010$.] (4 points)
- (d) **Prove** that the collection of Turing decidable languages is closed under intersection. (2 points)

Problem 6 (Bonus): Turing Machines (5 points)

Let $HALT_{TM} = \{\langle M, w \rangle : M \text{ is a TM and } M \text{ halts on input } w\}$. Given that $HALT_{TM}$ is undecidable, **prove** that the collection of Turing recognizable languages is not closed under complement.



After you are finished with the exam, please indicate where you stand on the smiley face spectrum.

